

MID SEM 2: MTH201

Name:

Reg No.:

Time: 1 h Marks: 20

Q 1. Consider the function $f(x) = x \log x$.

- (i) What is the largest domain of f as a subset of $\mathbb{R}$? [1]
- (ii) Calculate $\lim_{x \rightarrow 0^+} f(x)$. [2]
- (iii) Find the largest intervals where f is increasing. [1]
- (iv) Find the points of inflection of the function. [1]

Q 2. Calculate the Taylor polynomial of $f(x) = \sin(3x)$ as a polynomial of degree 5 around the point 0. [4]

Q 3. (i) Let $f(x) = x^2$. For the interval $[0, 3]$, consider the partition $P = \{0, 1, 3/2, 3\}$. Calculate the upper sum $U(P, f)$ and the lower sum $L(P, f)$. [2.5]

(ii) Calculate $\int_{-\pi/2}^{\pi} (|x| + |\cos(x)|) dx$. [2.5]

(iii) Let $F(x) = \int_0^x f(t) dt$ for an integrable function f . Calculate $F'(x)$. [2]

Q 4. State True or False with a short proof.

- (i) Let f be a function on a subset D of $\mathbb{R}$ such that $f'(x) = 0$ for all $x \in D$. Then f is a constant function. [1]
- (ii) All continuous functions on an open interval have a local maximum and minimum. [1]
- (iii) If c is a point of minimum of a continuous function $f(x)$ on a closed interval, then f is differentiable at c and $f'(c) = 0$. [1]
- (iv) All functions on the closed interval $[0, 1]$ are integrable. [1]